

ODD 2024
Tutorial Sheet– 2

Software Development Fundamentals – I (15B11CI111)

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration, and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real-world problems	Apply (Level 3)

Note: Students are advised to submit their solutions to their respective tutorial faculties

1. [CO2] What is the correct value to return to the operating system upon the successful completion of a program?

- A. -1
- B. 1
- C. 0
- D. Programs do not return a value.

2. [CO2] What is the only function all C++ programs must contain?

- A. start()
- B. system()
- C. main()
- D. program()

3. [CO2] What punctuation is used to signal the beginning and end of code blocks?

- A. { }
- B. -> and <-
- C. BEGIN and END
- D. (and)

4. [CO2] What punctuation ends most lines of C++ code?

- A. .
- B. ;
- C. :
- D. '

5. [CO2] Which of the following is a correct comment?

- A. /* Comments */
- B. ** Comment **
- C. /* Comment */
- D. { Comment }

6. [CO2] Which of the following is not a correct variable type?

- A. float
- B. real
- C. int
- D. double

7. Which of the following is the correct operator to compare two variables?

- A. :=

- B. =
- C. equal
- D. ==

Q8. [CO2] Identify & correct errors in following statements:

- a. int a=10, int b=20;
- b. int a=10, float b=2.5;
- c. const int number = 100;
number=500;
- d. Printf("To err is human");
- e. Printf(1,2,3);
- f. int b = 2f;

Q9. [CO2] What will be the output of following C programs? Give explanation.

```
(a)#include<stdio.h>
main()
{
int a,b ;
printf("Enter two numbers");
scanf("%d%d",&a,&b);
printf("%d + %d = %d\n",a,b,a+b);
printf("%d / %d = %d\n",a,b,a/b);
printf("%d * %d = %d\n",a,b,a*b);
printf("%d %% %d = %d\n",a,b,a%b);
}
```

```
(b)#include<stdio.h>
main()
{
printf("%d\t%d\t%d\t%d\t%d\n", 72, 072,
0072, 0x72, 0X72);
printf("%d\t%c\n",'A','A');
printf("%d\t%o\t%x\n",72,72,72);
printf("%i\t%i\t%i\t%i\n", 72, 072, 0x72,
0X72);
```

```
printf("%d\t%c");
}
```

Q10. [CO2] Give the output of following print statements:

- (a) printf("%05d,%5d,%-5d", 32,32,32);
- (b) printf("%.3f,%06.3f,%09.3f,%-09.3f,%6.0f,%6.0f", 45.6, 45.6, 45.6, 45.6, 45.4, 45.6);
- (c) printf("int: %d, char: %c, double: %lf\n", 5, '*', 5.0);
- (d) printf("you got %d%% in the exam\n", 100);

Q11. [CO2] Give the output of following program.

```
#include<stdio.h>
int main()
{char ch;
int i;
scanf("%c", &i);
scanf("%d", &ch);
printf("%c %d", ch, i);
return 0;
}
```

Q12[CO2] Give the output of following program.

```
#include<stdio.h>
int main()
{
float fval=7.29;
printf("%d\n", (int)fval);
return 0;
}
```